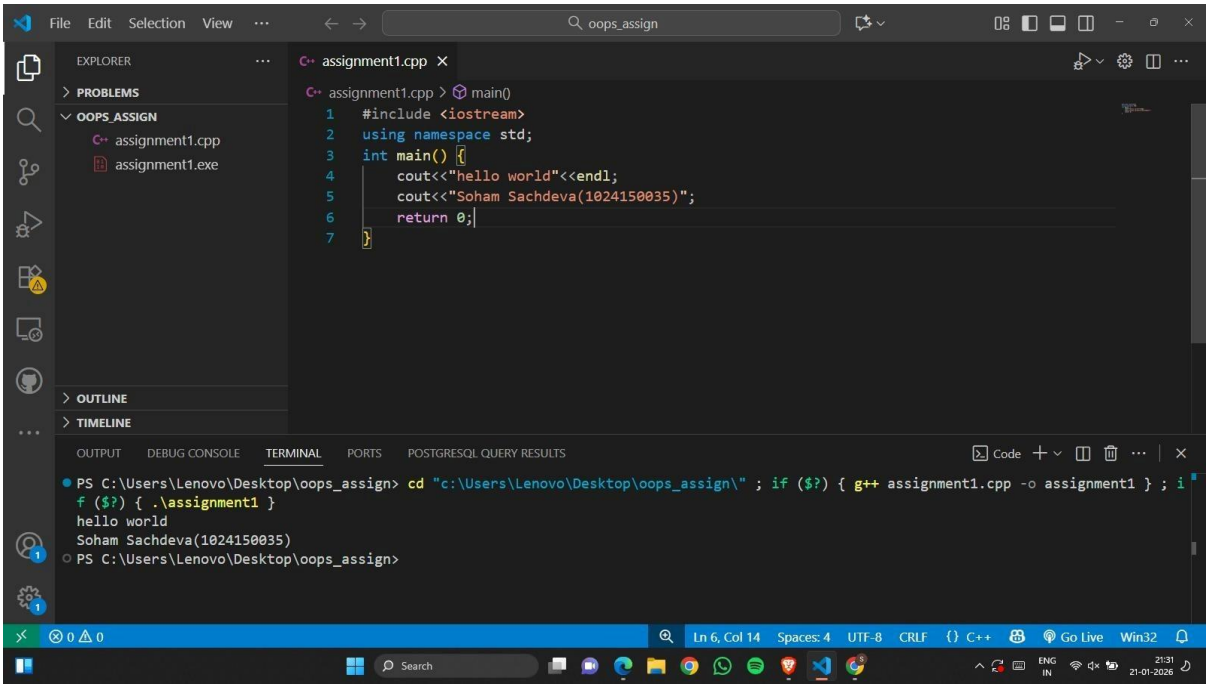
OOPS ASSIGNMENT-1

SOHAM SACHDEVA

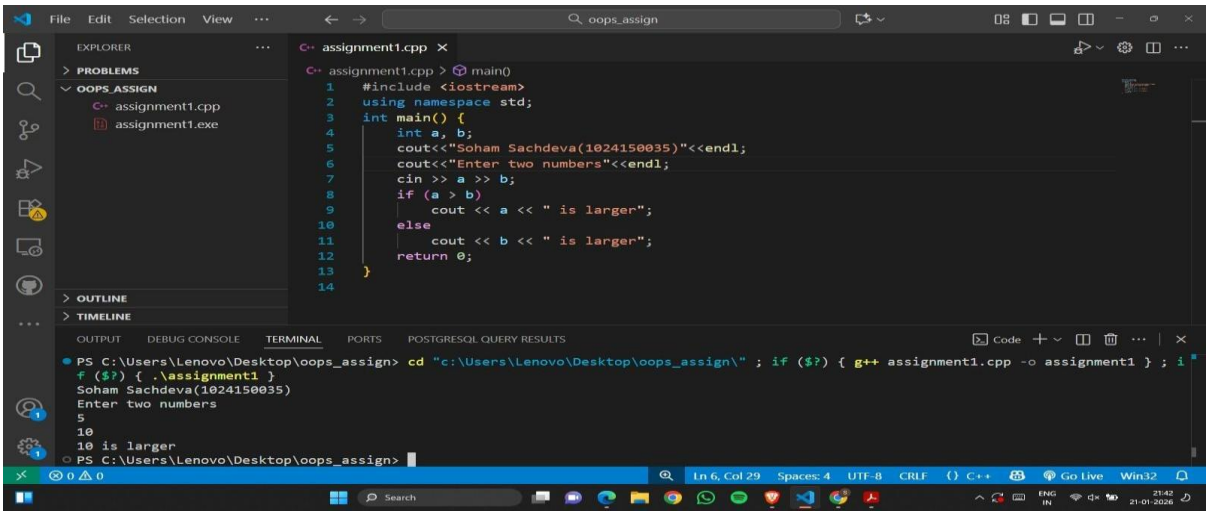
1024150035

2O11

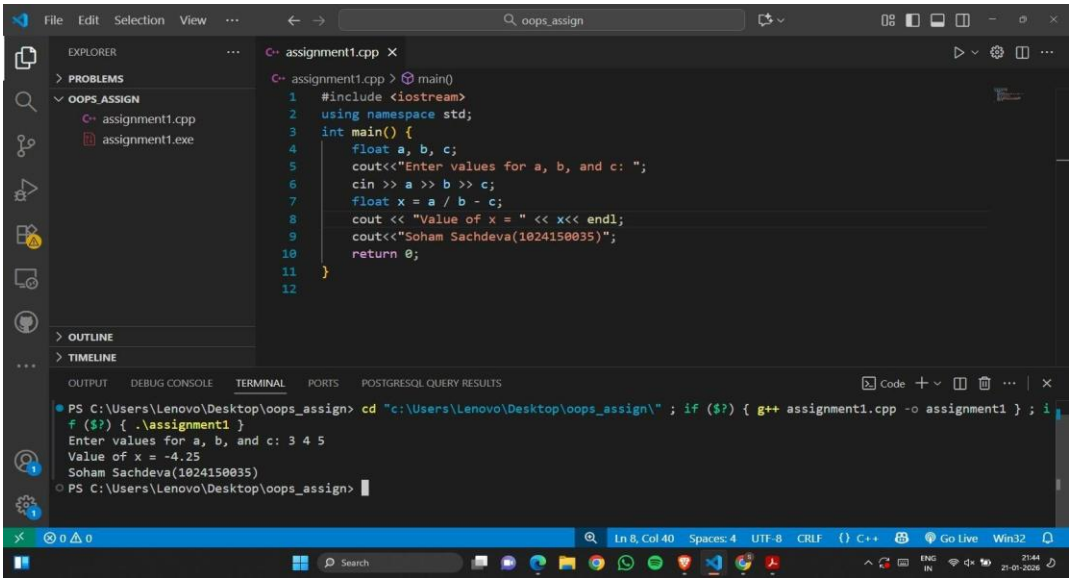
1. Write a program (WAP) to display "Hello World" on console display.



2. WAP to that reads two values from keyboard and displays the larger value on the screen.



3. WAP to read a, b, c and calculate x = a/b – c



4. WAP to implement the following control characters: -

‘\n’ is for new line, or you can use endl –

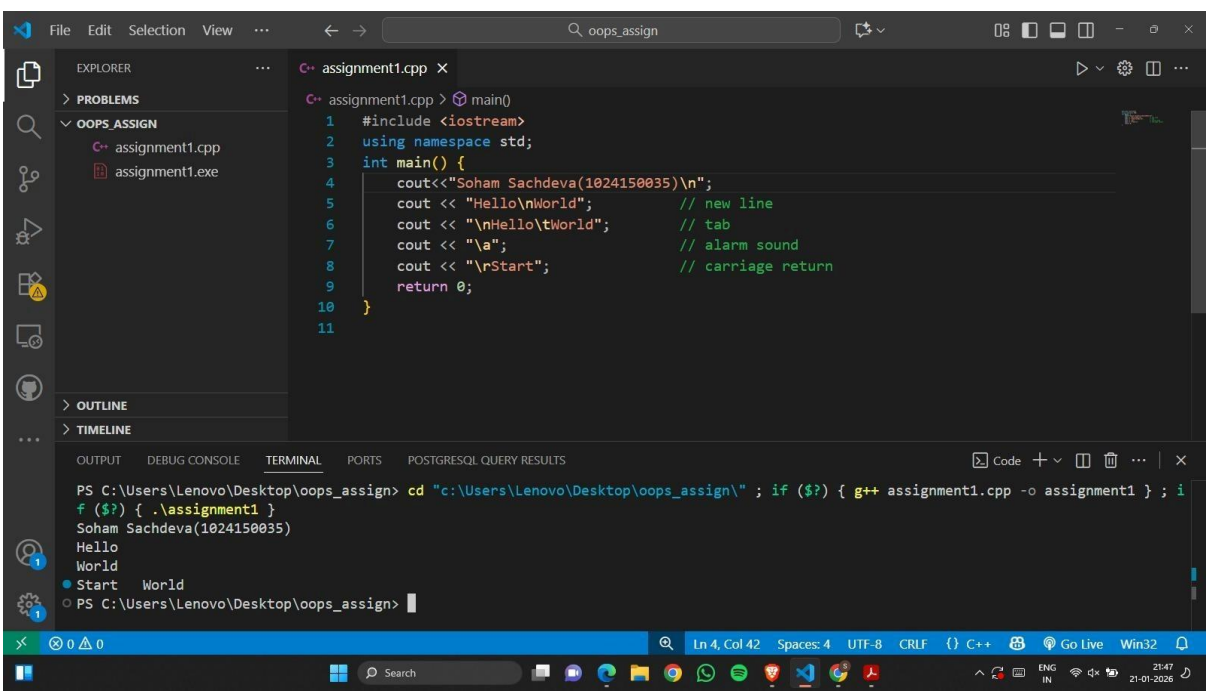
cout<<endl<<“message”; -

‘\t’ is for tab

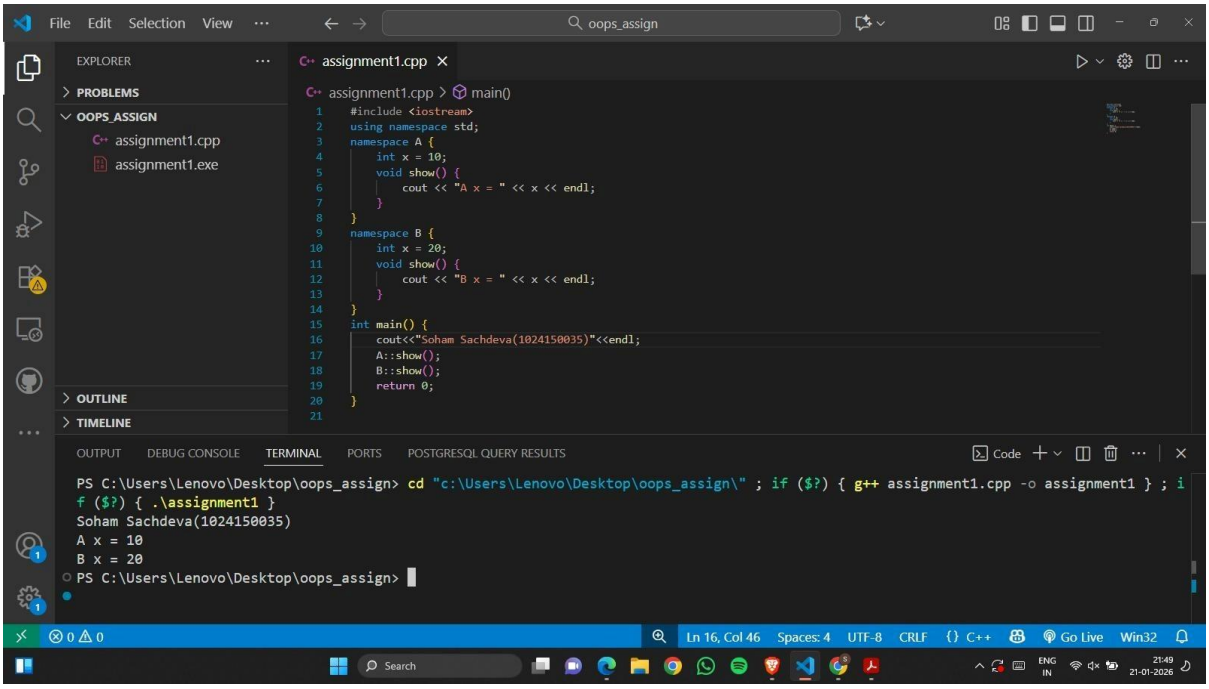
‘\a’ is an alarm sound

‘\r’ is carriage return to go to the beginning of the

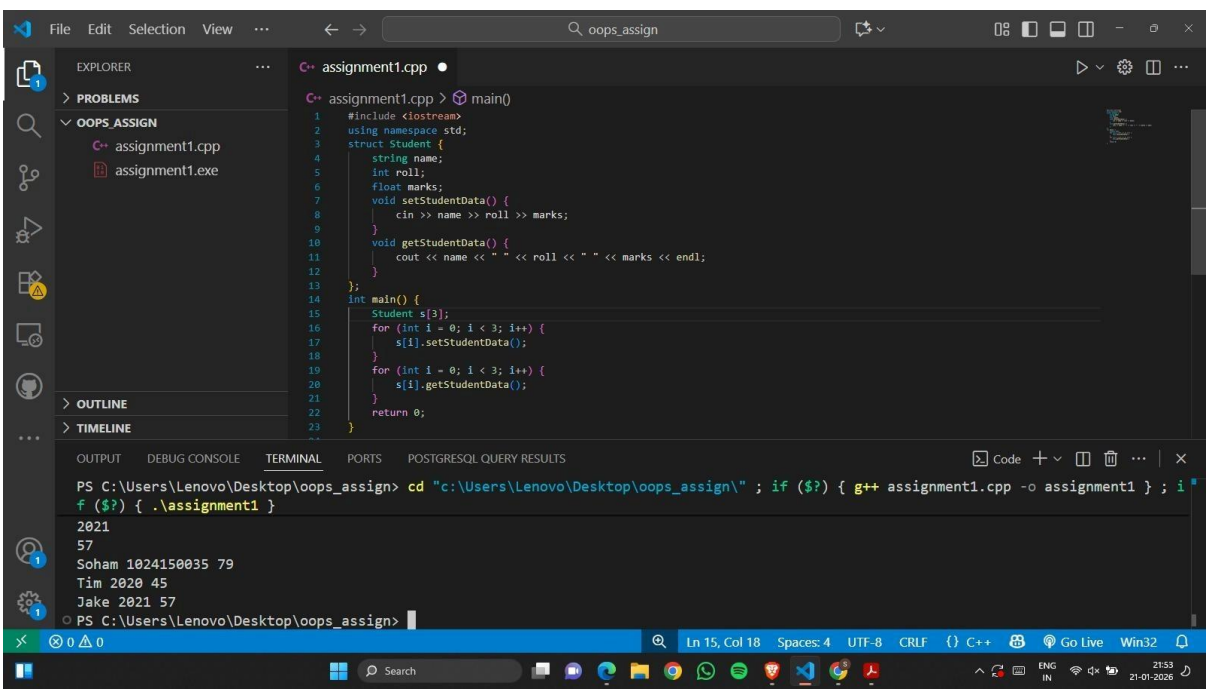
current line



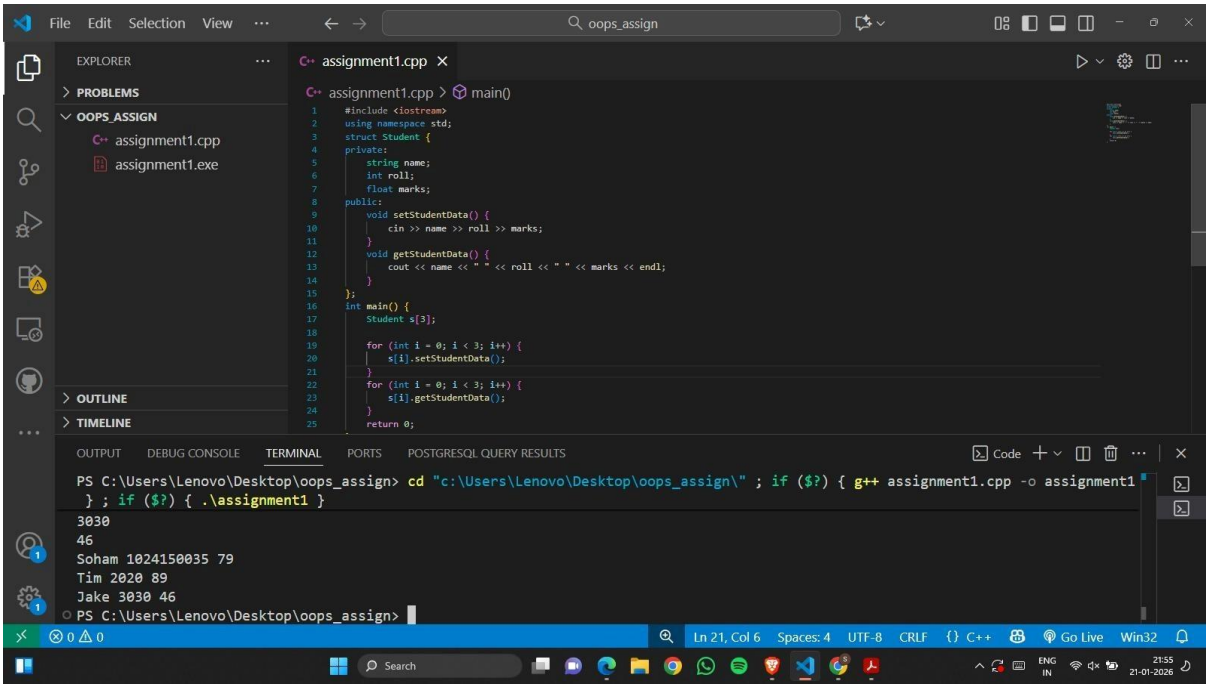
5. Implement namespace in a program to illustrate the use of same name variables and functions in different sections/libraries of the code.



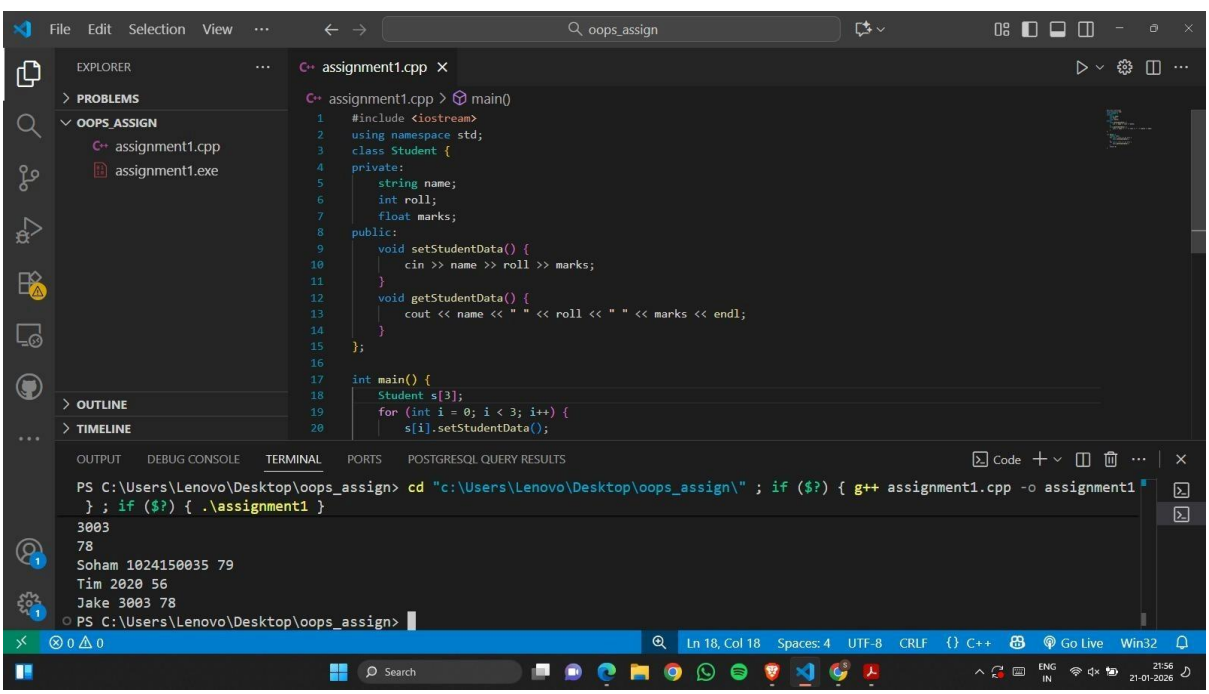
6. Write a program to define a structure Student that contains 3 variables and 2 functions: - Variables: (a) name (b) roll (c) marks - Functions: (1) setStudentData() (2) getStudentData() Use loops to input and output data for 3 student object entries.



7. In the previous program make the data private, and functions public.



8. Convert program #4 from struct to class.

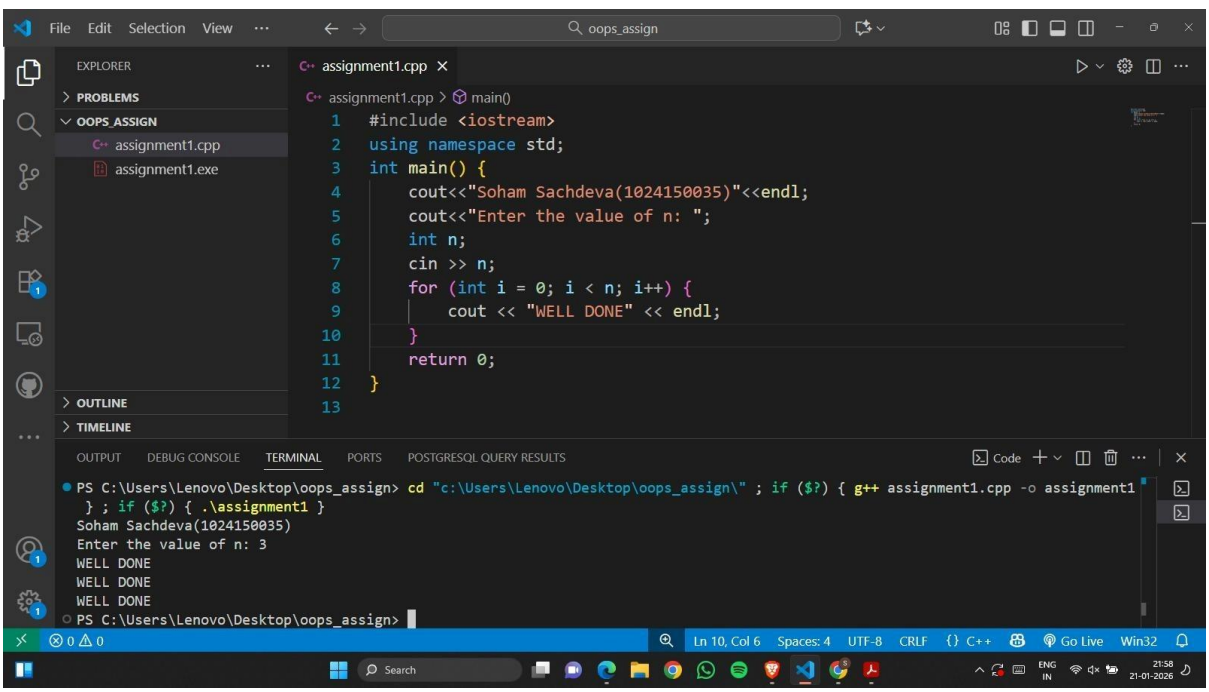


9. Discuss the differences of structures in C and structures in C++ during the lab with other students or your instructor.

ANSWER:

- In C, structure can store only data.
- In C++, structure can store data and functions.
- Access specifiers are not available in C.
- Access specifiers (public/private) are available in C++.
- C++ structure supports object-oriented programming

10. WAP to read an integer value from the keyboard and display on screen “WELL DONE”



11. WAP to use assignment operators += and -=

```cpp
#include <iostream>
using namespace std;
int main() {
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    int a = 10;
    a += 5;   // a = a + 5
    cout << "After += 5, a = " << a << endl;
    a -= 3;   // a = a - 3
    cout << "After -= 3, a = " << a << endl;
    return 0;
}
```

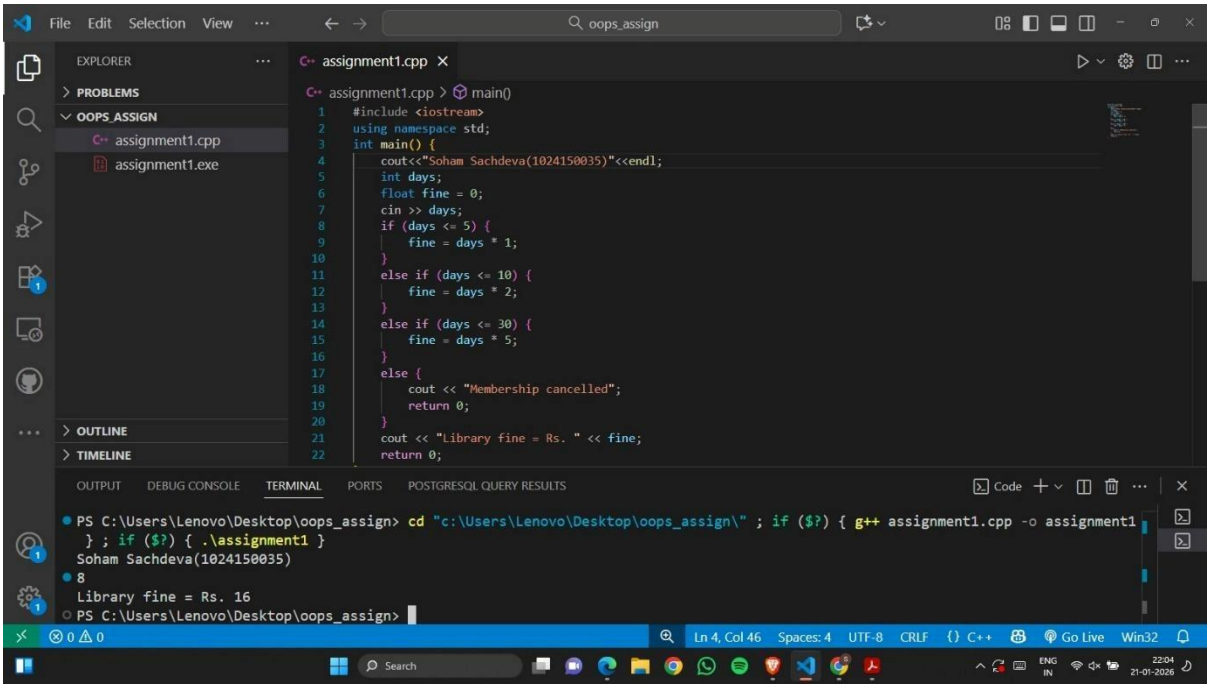
```
PS C:\Users\Lenovo\Desktop\oops_assign> cd "c:\Users\Lenovo\Desktop\oops_assign\" ; if ($?) { g++ assignment1.cpp -o assignment1
 } ; if ($?) { .\assignment1 }
Soham Sachdeva(1024150035)
After += 5, a = 15
After -= 3, a = 12
PS C:\Users\Lenovo\Desktop\oops_assign>
```

12.WAP to swap two numbers using bitwise operator

```cpp
#include <iostream>
using namespace std;
int main() {
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    int a, b;
    cin >> a >> b;
    a = a ^ b;
    b = a ^ b;
    a = a ^ b;
    cout << "After swapping:\n";
    cout << "a = " << a << endl;
    cout << "b = " << b << endl;
    return 0;
}
```

```
PS C:\Users\Lenovo\Desktop\oops_assign> cd "c:\Users\Lenovo\Desktop\oops_assign\" ; if ($?) { g++ assignment1.cpp -o assignment1
 } ; if ($?) { .\assignment1 }
Soham Sachdeva(1024150035)
4
8
After swapping:
a = 8
b = 4
```

13.Library fine calculation program (simple if-else)



OOPS ASSIGNMENT-2

SOHAM SACHDEVA

1024150035

2O11

QUES-1

main.cpp

```cpp
#include <iostream>
using namespace std;
struct Student {
    string name;
    int rollNo;
    string degree;
    string hostel;
    float cgpa;
    void addDetails() {
        cout << "Enter Name: ";
        cin >> name;
        cout << "Enter Roll No: ";
        cin >> rollNo;
        cout << "Enter Degree: ";
        cin >> degree;
        cout << "Enter Hostel: ";
        cin >> hostel;
        cout << "Enter CGPA: ";
        cin >> cgpa;
    }
    void updateCGPA(float newCGPA) {
        cgpa = newCGPA;
    }
    void updateHostel(string newHostel) {
        hostel = newHostel;
    }

    void displayDetails() {
        cout << "\nName: " << name
             << "\nRoll No: " << rollNo
             << "\nDegree: " << degree
             << "\nHostel: " << hostel
             << "\nCGPA: " << cgpa << endl;
    }
};
int main() {
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    Student s;
    s.addDetails();
    s.updateCGPA(9);
    s.updateHostel("Neeram Hall");
    s.displayDetails();
}
```

Output

```
Soham Sachdeva(1024150035)
Enter Name: Sam
Enter Roll No: 23
Enter Degree: Btech
Enter Hostel: D
Enter CGPA: 8

Name: Sam
Roll No: 23
Degree: Btech
Hostel: Neeram Hall
CGPA: 9


=== Code Execution Successful ===
```

QUES-2

main.cpp

```cpp
#include <iostream>
using namespace std;
class Student {
private:
    string name;
    int rollNo;
    float cgpa;

    void updateCGPA(float c) {
        cgpa = c;
    }
public:
    void addDetails() {
        cout << "Enter Name, Roll No, CGPA: ";
        cin >> name >> rollNo >> cgpa;
    }
    void modifyCGPA(float c) {
        updateCGPA(c);
    }
    void display() {
        cout << name << " " << rollNo << " " << cgpa << endl;
    }
};
int main() {
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    Student s;
    s.addDetails();
    s.modifyCGPA(8.5);
    s.display();
}
```

Output

```
Soham Sachdeva(1024150035)
Enter Name, Roll No, CGPA: Soham
1024150035
10
Soham 1024150035 8.5


=== Code Execution Successful ===
```

QUES-3

main.cpp

```cpp
#include <iostream>
using namespace std;
class Demo {
private:
    void secret() {
        cout << "Private function called by Soham Sachdeva(1024150035)\n";
    }
public:
    void show() {
        secret();
    }
};
int main() {

    Demo d;
    d.show();
}
```

Output

```
Private function called by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-4

main.cpp

```cpp
#include <iostream>
using namespace std;
class Rectangle {
int width, height;
public:
    void getdata() {
        cout << "Enter width: ";
        cin >> width;
        cout << "Enter height: ";
        cin >> height;
    }
    void calculatearea() {
        cout << "Area = " << width * height << endl;
    }
};
int main() {
    Rectangle r;
    r.getdata();
    r.calculatearea();
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
}
```

Output

```
Enter width: 5
Enter height: 10
Area = 50
Done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-5

main.cpp

```cpp
#include <iostream>
using namespace std;
class Complex {
    float real, imag;
public:
    void setComplex(float r, float i) {
        real = r;
        imag = i;
    }
    void displayComplex() {
        cout << real << " + " << imag << "i\n";
    }
    Complex add(Complex c) {
        Complex temp;
        temp.real = real + c.real;
        temp.imag = imag + c.imag;
        return temp;
    }
};
int main() {
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    Complex c1, c2, sum;
    c1.setComplex(2, 5);
    c2.setComplex(4, 6);
    sum = c1.add(c2);
    sum.displayComplex();
}
```

Output

```
Soham Sachdeva(1024150035)
6 + 11i


=== Code Execution Successful ===
```

QUES-6(a)

main.cpp

```cpp
// question 6 part a
#include <iostream>
using namespace std;
class Test {
public:
    void show();
};
void Test::show() {
    cout << "Function defined outside class" << endl;
}
int main() {
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    Test t;
    t.show();
    return 0;
}
```

Output

```
Soham Sachdeva(1024150035)
Function defined outside class


=== Code Execution Successful ===
```

QUES-6(b)

main.cpp

```cpp
// question 6 part b
#include <iostream>
using namespace std;
int x = 10;
int main() {
    int x = 20;
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    cout << "Local x: " << x << endl;
    cout << "Global x: " << ::x << endl;
    return 0;
}

```

Output

```
Soham Sachdeva(1024150035)
Local x: 20
Global x: 10


=== Code Execution Successful ===
```

QUES-6(c)

main.cpp

```cpp
// question 6 part c
#include <iostream>
using namespace std;
class variable {
public:
    static int x;
};
int variable::x = 50;
int main() {
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    cout << variable::x << endl;
    return 0;
}
```

Output

```
Soham Sachdeva(1024150035)
50


=== Code Execution Successful ===
```

QUES-6(d)

main.cpp

```cpp
// question 6 part c
#include <iostream>
using namespace std;
class variable {
public:
    static int x;
};
int variable::x = 50;
int main() {
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    cout << variable::x << endl;
    return 0;
}
```

Output

```
Soham Sachdeva(1024150035)
50


=== Code Execution Successful ===
```

QUES-7

main.cpp

```cpp
#include <iostream>
using namespace std;
namespace A {
    int x = 10;
    void show() {
        cout << "Namespace A\n";
    }
}
namespace B {
    int x = 20;
    void show() {
        cout << "Namespace B\n";
    }
}
int main() {
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    cout << A::x << endl;
    cout << B::x << endl;
    A::show();
    B::show();
}
```

Output

```
Soham Sachdeva(1024150035)
10
20
Namespace A
Namespace B


=== Code Execution Successful ===
```

OOPS ASSIGNMENT-3

SOHAM SACHDEVA

1024150035

2O11

QUES-1

main.cpp

```cpp
#include <iostream>
using namespace std;

class Student {
    int marks;
public:
    void set(int marks) {
        this->marks = marks;
    }
    void show() {
        cout << "Marks: " << marks << endl;
    }
};

int main() {
    Student s;
    Student *ptr = &s;
    s.set(90);
    s.show();
    ptr->set(95);
    ptr->show();
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Marks: 90
Marks: 95
Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-2

main.cpp

```cpp
#include <iostream>
using namespace std;
class B;
class A {
    int x;
public:
    A(int x) { this->x = x; }
    friend void swap(A &, B &);
    void show() { cout << "A: " << x << endl; }
};
class B {
    int y;
public:
    B(int y) { this->y = y; }
    friend void swap(A &, B &);
    void show() { cout << "B: " << y << endl; }
};
void swap(A &a, B &b) {
    int temp = a.x;
    a.x = b.y;
    b.y = temp;
}

int main() {
    A a(10);
    B b(20);
    swap(a, b);
    a.show();
    b.show();
    cout<<"Soham Sachdeva(1024150035)"<<endl;
}
```

Output

```
A: 20
B: 10
Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-3

main.cpp

```cpp
#include <iostream>
using namespace std;

class B;

class A {
    int x;
public:
    A(int x) { this->x = x; }
    friend int add(A, B);
};

class B {
    int y;
public:
    B(int y) { this->y = y; }
    friend int add(A, B);
};

int add(A a, B b) {
    return a.x + b.y;
}

int main() {
    A a(5);
    B b(15);
    cout << "Sum = " << add(a, b);
    cout<<"Soham Sachdeva(1024150035)"<<endl;
}
```

Output

```
Sum = 20Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-4

main.cpp

```cpp
#include <iostream>
using namespace std;

class A {
    int x = 100;
    friend class B;
};

class B {
public:
    void show(A a) {
        cout << "Value of x: " << a.x << endl;
    }
};

int main() {
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    A a;
    B b;
    b.show(a);
}
```

Output

```
Soham Sachdeva(1024150035)
Value of x: 100


=== Code Execution Successful ===
```

QUES-5

main.cpp

```cpp
#include <iostream>
using namespace std;

class Rectangle {
    int l, b;
public:
    void set() {
        cin >> l >> b;
    }
    void area() {
        cout << "Area = " << l * b << endl;
    }
};

int main() {
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    Rectangle r[3];

    for(int i = 0; i < 3; i++)
        r[i].set();

    for(int i = 0; i < 3; i++)
        r[i].area();
}
```

Output

```
Soham Sachdeva(1024150035)
5
4
3
21
1
7
Area = 20
Area = 63
Area = 7


=== Code Execution Successful ===
```

QUES-6

main.cpp

```cpp
#include <iostream>
using namespace std;

inline int cube(int x) {
    return x * x * x;
}

int main() {
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    int n;
    cin >> n;
    cout << "Cube = " << cube(n);
}
```

Output

```
Soham Sachdeva(1024150035)
4
Cube = 64

=== Code Execution Successful ===
```

QUES-7(a)

main.cpp

```cpp
// part A
#include <iostream>
using namespace std;

class Number {
public:
    int x;
    Number(int x) { this->x = x; }
};

Number modify(Number n) {
    n.x += 10;
    return n;
}

int main() {
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    Number n1(5);
    Number n2 = modify(n1);

    cout << n2.x;
}
```

Output

```
Soham Sachdeva(1024150035)
15

=== Code Execution Successful ==
```

QUES-7(b)

main.cpp

```cpp
// part B
#include <iostream>
using namespace std;

class Number {
public:
    int x;
    Number(int x) { this->x = x; }
};

Number modify(Number *n) {
    n->x += 10;
    return *n;
}

int main() {
    cout<<"Soham Sachdeva(1024150035)"<<endl;
    Number n1(5);
    Number n2 = modify(&n1);

    cout << n2.x;
}
```

Output

```
Soham Sachdeva(1024150035)
15

=== Code Execution Successful ===
```

OOPS ASSIGNMENT-4

SOHAM SACHDEVA

1024150035

2O11

QUES-1

```cpp
#include <iostream>
using namespace std;
class Rectangle {
    float length, breadth;
public:
    Rectangle() {
        length = 0;
        breadth = 0;
        cout << "Default constructor called (length = 0, breadth = 0)\n";
    }
    Rectangle(float x) {
        length = x;
        breadth = x;
        cout << "One parameter constructor called\n";
        cout << "Length = Breadth = " << x << endl;
    }
    Rectangle(float l, float b) {
        length = l;
        breadth = b;
        cout << "Two parameter constructor called\n";
        cout << "Length = " << l << " Breadth = " << b << endl;
    }
    float area() {
        return length * breadth;
    }
};
int main() {
    cout << "Creating object r1 (no parameter)\n";
    Rectangle r1;
    cout << "Area of r1 = " << r1.area() << endl;
    cout << "\nCreating object r2 (one parameter)\n";
    float x;
    cout << "Enter value: ";
    cin >> x;
    Rectangle r2(x);
    cout << "Area of r2 = " << r2.area() << endl;
    cout << "\nCreating object r3 (two parameters)\n";
    float l, b;
    cout << "Enter length and breadth: ";
    cin >> l >> b;
    Rectangle r3(l, b);
    cout << "Area of r3 = " << r3.area() << endl;
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

```
Creating object r1 (no parameter)
Default constructor called (length = 0, breadth = 0)
Area of r1 = 0

Creating object r2 (one parameter)
Enter value: 4
One parameter constructor called
Length = Breadth = 4
Area of r2 = 16

Creating object r3 (two parameters)
Enter length and breadth: 4
5
Two parameter constructor called
Length = 4 Breadth = 5
Area of r3 = 20
Done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-2

main.cpp

```cpp
#include <iostream>
using namespace std;
class Rectangle {
    float length, breadth;
public:
    Rectangle(float l = 0, float b = 0) {
        length = l;
        breadth = b;
        cout << "Constructor called -> Length: " << l << " Breadth: " << b << endl;
    }
    float area() {
        return length * breadth;
    }
    ~Rectangle() {
        cout << "Destructor called\n";
    }
};
int main() {
    int n;
    cout << "Enter number of rectangles: ";
    cin >> n;
    Rectangle r[n];
    for (int i = 0; i < n; i++) {
        float l, b;
        cout << "\nEnter length and breadth for rectangle " << i + 1 << ": ";
        cin >> l >> b;
        r[i] = Rectangle(l, b);
    }
    cout << "\nAreas of rectangles:\n";
    for (int i = 0; i < n; i++) {
        cout << "Rectangle " << i + 1 << " Area = " << r[i].area() << endl;
    }
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter number of rectangles: 4
Constructor called -> Length: 0 Breadth: 0
Constructor called -> Length: 0 Breadth: 0
Constructor called -> Length: 0 Breadth: 0
Constructor called -> Length: 0 Breadth: 0

Enter length and breadth for rectangle 1: 2
2
Constructor called -> Length: 2 Breadth: 2
Destructor called

Enter length and breadth for rectangle 2: 5
6
Constructor called -> Length: 5 Breadth: 6
Destructor called

Enter length and breadth for rectangle 3: 7
9
Constructor called -> Length: 7 Breadth: 9
Destructor called

Enter length and breadth for rectangle 4: 7
3
Constructor called -> Length: 7 Breadth: 3
Destructor called

Areas of rectangles:
Rectangle 1 Area = 4
Rectangle 2 Area = 30
Rectangle 3 Area = 63
Rectangle 4 Area = 21
Done by Soham Sachdeva(1024150035)
Destructor called
Destructor called
Destructor called
Destructor called


=== Code Execution Successful ===
```

QUES-3

main.cpp

```cpp
#include <iostream>
using namespace std;
class Demo {
public:
    Demo() {
        cout << "Constructor called\n";
    }
    ~Demo() {
        cout << "Destructor called\n";
    }
};
int main() {
    cout << "Creating object d1\n";
    Demo d1;
    cout << "Creating object d2\n";
    Demo d2;
    cout << "End of main function\n";
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Creating object d1
Constructor called
Creating object d2
Constructor called
End of main function
Done by Soham Sachdeva(1024150035)
Destructor called
Destructor called


=== Code Execution Successful ===
```

QUES-4

main.cpp

```cpp
class Test {
public:
    int x;
    Test() {
        cout << "Constructor called\n";
    }
    ~Test() {
        cout << "Destructor called\n";
    }
};
int main() {
    int *p;
    p = new int;
    cout << "Enter integer value: ";
    cin >> *p;
    cout << "Value = " << *p << endl;
    delete p;
    float *f;
    f = new float;
    cout << "\nEnter float value: ";
    cin >> *f;
    cout << "Value = " << *f << endl;
    delete f;
    int n;
    cout << "\nEnter size of integer array: ";
    cin >> n;
    int *arr = new int[n];
    cout << "Enter elements:\n";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    cout << "Array elements:\n";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
    delete[] arr;
    Test *obj = new Test();
    delete obj;
    int m;
    cout << "\nEnter number of objects: ";
    cin >> m;
    Test *objArr = new Test[m];
    delete[] objArr;
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
```

Output

```
Enter integer value: 3
Value = 3

Enter float value: 3.3
Value = 3.3

Enter size of integer array: 3
Enter elements:
1

4
4
Array elements:
1 4 4
Constructor called
Destructor called

Enter number of objects: 3
Constructor called
Constructor called
Constructor called
Destructor called
Destructor called
Destructor called
Done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

OOPS ASSIGNMENT-5

SOHAM SACHDEVA

1024150035

2O11

QUES-1

main.cpp

```cpp
#include <iostream>
using namespace std;
class Base {
public:
    int x;
    void getBase() {
        cout << "Enter value of x: ";
        cin >> x;
    }
    void showBase() {
        cout << "Value of x = " << x << endl;
    }
};
class Derived : public Base {
public:
    int y;
    void getDerived() {
        cout << "Enter value of y: ";
        cin >> y;
    }
    void showDerived() {
        cout << "Value of y = " << y << endl;
    }
};
int main() {
    Base b;
    b.getBase();
    b.showBase();
    Derived d;
    d.getBase();
    d.getDerived();
    d.showBase();
    d.showDerived();
    cout<<"Done by Soham Sachdeva"<<endl;
    return 0;
}
```

Output

```
Enter value of x: 4
Value of x = 4
Enter value of x: 5
Enter value of y: 2
Value of x = 5
Value of y = 2
Done by Soham Sachdeva


=== Code Execution Successful ===
```

QUES-2

main.cpp

```cpp
#include <iostream>
using namespace std;
class Base {
protected:
    int x;
public:
    void getData() {
        cout << "Enter value: ";
        cin >> x;
    }
};
class Derived : public Base {
public:
    void show() {
        cout << "Accessing protected x = " << x << endl;
    }
};
int main() {
    Derived d;
    d.getData();
    d.show();
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter value: 4
Accessing protected x = 4
Done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-3

main.cpp

```cpp
#include <iostream>
using namespace std;
class Base {
public:
    int a;
protected:
    int b;
private:
    int c;
public:
    void getData() {
        cout << "Enter a, b, c: ";
        cin >> a >> b >> c;
    }
    void showBase() {
        cout << "a=" << a << " b=" << b << " c=" << c << endl;
    }
};
class Derived : public Base {
public:
    void show() {
        cout << "Public inheritance a=" << a << " b=" << b << endl;
    }
};
int main() {
    Derived d;
    d.getData();
    d.show();
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter a, b, c: 4 5 6
Public inheritance a=4 b=5
Done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-4(a)

main.cpp

```cpp
#include <iostream>
using namespace std;
class A {
public:
    void showA() {
        cout << "Class A\n";
    }
};
class B : public A {
public:
    void showB() {
        cout << "Class B\n";
    }
};
int main() {
    B obj;
    obj.showA();
    obj.showB();
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Class A
Class B
Done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-4(b)

main.cpp

```cpp
#include <iostream>
using namespace std;
class A {
public:
    void showA() {
        cout << "Class A\n";
    }
};
class B {
public:
    void showB() {
        cout << "Class B\n";
    }
};
class C : public A, public B {
public:
    void showC() {
        cout << "Class C\n";
    }
};
int main() {
    C obj;
    obj.showA();
    obj.showB();
    obj.showC();
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Class A
Class B
Class C
Done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-4(c)

main.cpp

```cpp
#include <iostream>
using namespace std;
class A {
public:
    void showA() {
        cout << "Class A\n";
    }
};
class B : public A {
public:
    void showB() {
        cout << "Class B\n";
    }
};
class C : public A {
public:
    void showC() {
        cout << "Class C\n";
    }
};
int main() {
    B b;
    C c;
    b.showA();
    b.showB();
    c.showA();
    c.showC();
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Class A
Class B
Class A
Class C
Done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-4(d)

main.cpp

```cpp
#include <iostream>
using namespace std;
class A {
public:
    void showA() {
        cout << "Class A\n";
    }
};
class B : public A {};
class C : public A {};
class D : public B, public C {
public:
    void showD() {
        cout << "Class D\n";
    }
};
int main() {
    D obj;
    obj.B::showA();
    obj.C::showA();
    obj.showD();
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Class A
Class A
Class D
Done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-4(e)

main.cpp

```cpp
#include <iostream>
using namespace std;
class A {
public:
    void showA() {
        cout << "Class A\n";
    }
};
class B : public A {
public:
    void showB() {
        cout << "Class B\n";
    }
};
class C : public B {
public:
    void showC() {
        cout << "Class C\n";
    }
};
int main() {
    C obj;
    obj.showA();
    obj.showB();
    obj.showC();
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Class A
Class B
Class C
Done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-5

main.cpp

```cpp
#include <iostream>
using namespace std;
class Base {
public:
    Base() {
        cout << "Base constructor\n";
    }
    ~Base() {
        cout << "Base destructor\n";
    }
};
class Derived : public Base {
public:
    Derived() {
        cout << "Derived constructor\n";
    }
    ~Derived() {
        cout << "Derived destructor\n";
    }
};
int main() {
    Derived d;
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Base constructor
Derived constructor
Done by Soham Sachdeva(1024150035)
Derived destructor
Base destructor


=== Code Execution Successful ===
```

QUES-6

main.cpp

```cpp
#include <iostream>
using namespace std;
class Book {
public:
    string title, author;
    float price;
    void getBook() {
        cout << "Enter title, author, price: ";
        cin >> title >> author >> price;
    }
    void showBook() {
        cout << title << " " << author << " " << price << endl;
    }
};
class Textbook : public Book {
public:
    string subject;
    void getText() {
        cout << "Enter subject: ";
        cin >> subject;
    }
    void showText() {
        showBook();
        cout << "Subject: " << subject << endl;
    }
};
int main() {
    Textbook t;
    t.getBook();
    t.getText();
    t.showText();
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter title, author, price: OOPS
John
500
Enter subject: oops
OOPS John 500
Subject: oops
Done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-7

main.cpp

```cpp
#include <iostream>
using namespace std;
class Speedometer {
public:
    int speed;
    void getSpeed() {
        cout << "Enter speed: ";
        cin >> speed;
    }
};
class FuelGauge {
public:
    int fuel;
    void getFuel() {
        cout << "Enter fuel level: ";
        cin >> fuel;
    }
};
class Thermometer {
public:
    int temp;
    void getTemp() {
        cout << "Enter temperature: ";
        cin >> temp;
    }
};
class CarDashboard : public Speedometer, public FuelGauge, public Thermometer {
public:
    void display() {
        cout << "Speed: " << speed << endl;
        cout << "Fuel: " << fuel << endl;
        cout << "Temperature: " << temp << endl;
    }
};
int main() {
    CarDashboard c;
    c.getSpeed();
    c.getFuel();
    c.getTemp();
    c.display();
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter speed: 90
Enter fuel level: 5
Enter temperature: 29
Speed: 90
Fuel: 5
Temperature: 29
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-8

main.cpp

```cpp
#include <iostream>
using namespace std;
class LibraryUser {
public:
    string name;
    int id;
    void getUser() {
        cout << "Enter name and id: ";
        cin >> name >> id;
    }
};
class Student : public LibraryUser {
public:
    int grade;
    void getStudent() {
        cout << "Enter grade: ";
        cin >> grade;
    }
    void show() {
        cout << name << " " << id << " Grade: " << grade << endl;
    }
};
class Teacher : public LibraryUser {
public:
    string dept;
    void getTeacher() {
        cout << "Enter department: ";
        cin >> dept;
    }
    void show() {
        cout << name << " " << id << " Dept: " << dept << endl;
    }
};
int main() {
    Student s;
    Teacher t;
    s.getUser();
    s.getStudent();
    s.show();
    t.getUser();
    t.getTeacher();
    t.show();
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter name and id: Soham 23
Enter grade: 2
Soham 23 Grade: 2
Enter name and id: John
21
Enter department: Computer
John 21 Dept: Computer
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-9

main.cpp

```cpp
#include <iostream>
using namespace std;
class Vehicle {
public:
    string make, model;
    int year;
    void getVehicle() {
        cout << "Enter make, model, year: ";
        cin >> make >> model >> year;
    }
};
class Truck : public Vehicle {
public:
    int load;
    void getTruck() {
        cout << "Enter load capacity: ";
        cin >> load;
    }
};
class RefrigeratedTruck : public Truck {
public:
    int temp;
    void getRefrigerated() {
        cout << "Enter temperature control: ";
        cin >> temp;
    }
    void show() {
        cout << make << " " << model << " " << year << endl;
        cout << "Load: " << load << " Temp: " << temp << endl;
    }
};
int main() {
    RefrigeratedTruck r;
    r.getVehicle();
    r.getTruck();
    r.getRefrigerated();
    r.show();
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter make, model, year: honda city 2006
Enter load capacity: 20
Enter temperature control: 45
honda city 2006
Load: 20 Temp: 45
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-10

main.cpp

```cpp
#include <iostream>
using namespace std;
class Person {
public:
    string name;
    void getPerson() {
        cout << "Enter name: ";
        cin >> name;
    }
};
class Staff : virtual public Person {
public:
    int emp_id;
    void getStaff() {
        cout << "Enter employee id: ";
        cin >> emp_id;
    }
};
class Student : virtual public Person {
public:
    int student_id;
    void getStudent() {
        cout << "Enter student id: ";
        cin >> student_id;
    }
};
class TeachingAssistant : public Staff, public Student {
public:
    void show() {
        cout << "Name: " << name << endl;
        cout << "Emp ID: " << emp_id << endl;
        cout << "Student ID: " << student_id << endl;
    }
};
int main() {
    TeachingAssistant t;
    t.getPerson();
    t.getStaff();
    t.getStudent();
    t.show();
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter name: john
Enter employee id: 23
Enter student id: 32
Name: john
Emp ID: 23
Student ID: 32
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

OOPS ASSIGNMENT-6

SOHAM SACHDEVA

1024150035

2O11

QUES-1

main.cpp

```cpp
#include <iostream>
using namespace std;
class Book {
public:
    string title, author;
    int ISBN;
};
class Library {
    Book books[100];
    int count;
public:
    Library() {
        count = 0;
    }
    bool addNewBook(string &title, string &author, int &ISBN) {
        books[count].title = title;
        books[count].author = author;
        books[count].ISBN = ISBN;
        count++;
        return true;
    }
    bool removeBooks(int &ISBN);
    void displayDetails() {
        cout << "\nBooks in library:\n";
        for (int i = 0; i < count; i++) {
            cout << books[i].title << " " << books[i].author << " " << books[i].ISBN << endl;
        }
    }
};
bool Library::removeBooks(int &ISBN) {
    for (int i = 0; i < count; i++) {
        if (books[i].ISBN == ISBN) {
            for (int j = i; j < count - 1; j++) {
                books[j] = books[j + 1];
            }
            count--;
            return true;
        }
    }
    return false;
}
int main() {
    Library lib;
    for (int i = 0; i < 5; i++) {
        string t, a;
        int id;
```

Share Run

Output

```
Enter title, author, ISBN: oops
joe
43
Enter title, author, ISBN: dsa
john
34
Enter title, author, ISBN: cn
jack
54
Enter title, author, ISBN: os
peter
12
Enter title, author, ISBN: dbms
andrew
76

Books in library:
oops joe 43
dsa john 34
cn jack 54
os peter 12
dbms andrew 76

Enter ISBN to remove:
```

QUES-2

main.cpp

```cpp
    string title, author;
    int ISBN;

    Book() {
        title = "";
        author = "";
        ISBN = 0;
    }
    Book(string t, string a, int i) {
        this->title = t;
        this->author = a;
        this->ISBN = i;
    }

    Book(const Book &b) {
        this->title = b.title;
        this->author = b.author;
        this->ISBN = b.ISBN;
    }
};

class Library {
    Book books[100];
    int count;

public:
    Library() {
        count = 0;
    }

    bool addNewBook(string t, string a, int i) {
        books[count] = Book(t, a, i);
        count++;
        return true;
    }

    bool removeBooks(int i) {
        for (int j = 0; j < count; j++) {
            if (books[j].ISBN == i) {
                for (int k = j; k < count - 1; k++) {
                    books[k] = books[k + 1];
                }
                count--;
                return true;
            }
        }
```

Output

```
Enter title, author, ISBN: oops
joe
43
Enter title, author, ISBN: dsa
john
54
Enter title, author, ISBN: os
jack
21
oops joe 43
dsa john 54
os jack 21
Enter ISBN to remove: 21
oops joe 43
dsa john 54
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-3

main.cpp

```cpp
#include <iostream>
using namespace std;
class Account {
    const long acc_no;
    long transaction_id;
    string type;
    double balance;
public:
    Account(long a) : acc_no(a) {
        balance = 0;
        transaction_id = 0;
    }
    long depositAmount(const long to, const long from, const double amount) {
        balance += amount;
        type = "credit";
        transaction_id++;
        return transaction_id;
    }
    long creditAmount(const long to, const long from, const double amount) {
        balance -= amount;
        type = "debit";
        transaction_id++;
        return transaction_id;
    }
    void displayDetails() const {
        cout << "Account: " << acc_no << endl;
        cout << "Balance: " << balance << endl;
        cout << "Last Transaction: " << type << " ID: " << transaction_id << endl;
    }
};
```

Output

```
Account: 101
Balance: 300
Last Transaction: debit ID: 2
Account: 102
Balance: 1000
Last Transaction: credit ID: 1
Account: 103
Balance: 0
Last Transaction:  ID: 0
Account: 104
Balance: 0
Last Transaction:  ID: 0
Account: 105
Balance: 0
Last Transaction:  ID: 0
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-4

main.cpp

```cpp
#include <iostream>
using namespace std;
class B;
class A {
    int x;
public:
    void get() {
        cout << "Enter value for A: ";
        cin >> x;
    }
    friend void add(A, B);
};
class B {
    int y;
public:
    void get() {
        cout << "Enter value for B: ";
        cin >> y;
    }
    friend void add(A, B);
};
void add(A a, B b) {
    cout << "Sum = " << a.x + b.y << endl;
}
```

Output

```
Enter value for A: 4
Enter value for B: 6
Sum = 10
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-5

main.cpp

```cpp
#include <iostream>
using namespace std;
class Complex {
    int real, imag;
public:
    Complex(int r = 0, int i = 0) {
        real = r;
        imag = i;
    }
    Complex(const Complex &c) {
        real = c.real;
        imag = c.imag;
    }
    void display() {
        cout << real << " + " << imag << "i" << endl;
    }
    friend void sum(Complex, Complex);
};
void sum(Complex c1, Complex c2) {
    Complex c3;
    c3.real = c1.real + c2.real;
    c3.imag = c1.imag + c2.imag;
    cout << "Sum = ";
    c3.display();
}
int main() {
    int r1, i1, r2, i2;
    cout << "Enter first complex number: ";
    cin >> r1 >> i1;
    cout << "Enter second complex number: ";
    cin >> r2 >> i2;
    Complex c1(r1, i1), c2(r2, i2);
    sum(c1, c2);
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter first complex number: 4
3
Enter second complex number: 4
1
Sum = 8 + 4i
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

OOPS ASSIGNMENT-7

SOHAM SACHDEVA

1024150035

2O11

QUES-1

main.cpp

```cpp
#include <iostream>
using namespace std;
class Polygon {
protected:
    float width, height;
public:
    void set_value(float w, float h) {
        width = w;
        height = h;
    }
    virtual void area() {}
};
class Rectangle : public Polygon {
public:
    void area() {
        cout << "Rectangle Area = " << width * height << endl;
    }
};
class Triangle : public Polygon {
public:
    void area() {
        cout << "Triangle Area = " << 0.5 * width * height << endl;
    }
};
```

Output

```
Enter width and height: 4
10
Rectangle Area = 40
Triangle Area = 20
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-2

main.cpp

```cpp
public:
    virtual void area() {}
    virtual void display() {}
};
class Circle : public Shape {
    float r;
public:
    void get() {
        cout << "Enter radius: ";
        cin >> r;
    }
    void area() {
        cout << "Circle Area = " << 3.14 * r * r << endl;
    }
    void display() {
        cout << "Circle\n";
    }
};
class Rectangle : public Shape {
    float l, b;
public:
    void get() {
        cout << "Enter length and breadth: ";
        cin >> l >> b;
    }
    void area() {
        cout << "Rectangle Area = " << l * b << endl;
    }
    void display() {
        cout << "Rectangle\n";
    }
};
class Triangle : public Shape {
    float b, h;
public:
    void get() {
        cout << "Enter base and height: ";
        cin >> b >> h;
    }
    void area() {
        cout << "Triangle Area = " << 0.5 * b * h << endl;
    }
    void display() {
        cout << "Triangle\n";
    }
};
```

Output

```
Enter radius: 6
Circle
Circle Area = 113.04
Enter length and breadth: 4
2
Rectangle
Rectangle Area = 8
Enter base and height: 7
10
Triangle
Triangle Area = 35
done by Soham Sachdeva


=== Code Execution Successful ===
```

QUES-3

main.cpp

```cpp
#include <iostream>
#include<math.h>
using namespace std;
float area(float b, float h) {
    return 0.5 * b * h;
}
float area(float a) {
    return (1.732 / 4) * a * a;
}
float area(float a, float b, float c) {
    float s = (a + b + c) / 2;
    return sqrt(s * (s - a) * (s - b) * (s - c));
}
int main() {
    float b, h, a, x, y, z;
    cout << "Enter base and height: ";
    cin >> b >> h;
    cout << "Right Triangle Area = " << area(b, h) << endl;
    cout << "Enter side of equilateral triangle: ";
    cin >> a;
    cout << "Equilateral Area = " << area(a) << endl;
    cout << "Enter 3 sides: ";
    cin >> x >> y >> z;
    cout << "Isosceles Area = " << area(x, y, z) << endl;
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter base and height: 4
5
Right Triangle Area = 10
Enter side of equilateral triangle: 6
Equilateral Area = 15.588
Enter 3 sides: 1

3
3
Isosceles Area = 1.47902
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-4

main.cpp

```cpp
#include <iostream>
using namespace std;
class Student {
public:
    virtual void show() = 0;
};
class Engineering : public Student {
public:
    void show() {
        cout << "Engineering Student\n";
    }
};
class Medicine : public Student {
public:
    void show() {
        cout << "Medicine Student\n";
    }
};
class Science : public Student {
public:
    void show() {
        cout << "Science Student\n";
    }
};
int main() {
    Student *s[3];
    Engineering e;
    Medicine m;
    Science sc;
    s[0] = &e;
    s[1] = &m;
    s[2] = &sc;
    for (int i = 0; i < 3; i++) {
        s[i]->show();
    }
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Engineering Student
Medicine Student
Science Student
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-5

main.cpp

```cpp
#include <iostream>
using namespace std;
class Time {
    int h, m, s;
public:
    Time(int h1 = 0, int m1 = 0, int s1 = 0) {
        h = h1;
        m = m1;
        s = s1;
    }
    Time operator+(Time t) {
        Time temp;
        temp.s = s + t.s;
        temp.m = m + t.m + temp.s / 60;
        temp.s %= 60;
        temp.h = h + t.h + temp.m / 60;
        temp.m %= 60;
        return temp;
    }
    void show() {
        cout << h << ":" << m << ":" << s << endl;
    }
};
int main() {
    Time t1(5,15,34), t2(9,53,58), t3;
    t3 = t1 + t2;
    t3.show();
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
15:9:32
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-6

main.cpp

```cpp
#include <iostream>
using namespace std;
class STRING {
    string str;
public:
    void get() {
        cout << "Enter string: ";
        cin >> str;
    }
    bool operator==(STRING s) {
        return str == s.str;
    }
    STRING operator+(STRING s) {
        STRING temp;
        temp.str = str + s.str;
        return temp;
    }
    void show() {
        cout << str << endl;
    }
};
int main() {
    STRING s1, s2, s3;
    s1.get();
    s2.get();
    if (s1 == s2)
        cout << "Strings are equal\n";
    else
        cout << "Not equal\n";
    s3 = s1 + s2;
    cout << "Concatenated: ";
    s3.show();
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter string: soham
Enter string: sachdeva
Not equal
Concatenated: sohamsachdeva
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-7

main.cpp

```cpp
#include <iostream>
using namespace std;
class Matrix {
    int a[2][2];
public:
    void get() {
        cout << "Enter 4 elements: ";
        for (int i=0;i<2;i++)
            for (int j=0;j<2;j++)
                cin >> a[i][j];
    }
    void show() {
        for (int i=0;i<2;i++) {
            for (int j=0;j<2;j++)
                cout << a[i][j] << " ";
            cout << endl;
        }
    }
    friend Matrix operator*(Matrix, Matrix);
};
Matrix operator*(Matrix m1, Matrix m2) {
    Matrix m3;
    for (int i=0;i<2;i++)
        for (int j=0;j<2;j++) {
            m3.a[i][j]=0;
            for (int k=0;k<2;k++)
                m3.a[i][j]+=m1.a[i][k]*m2.a[k][j];
        }
    return m3;
}
int main() {
    Matrix m1, m2, m3;
    m1.get();
    m2.get();
    m3 = m1 * m2;
    cout << "Result:\n";
    m3.show();
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter 4 elements: 4
2
1
3
Enter 4 elements: 4
3
2
6
Result:
20 24
10 21
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-8

main.cpp

```cpp
#include <iostream>
using namespace std;
class Array {
    int a[5];
public:
    void get() {
        cout << "Enter 5 elements: ";
        for (int i=0;i<5;i++) cin >> a[i];
    }
    int operator[](int i) {
        if (i < 0 || i >= 5) {
            cout << "Index out of bounds\n";
            return -1;
        }
        return a[i];
    }
};
int main() {
    Array arr;
    arr.get();
    int index;
    cout << "Enter index: ";
    cin >> index;
    cout << "Value: " << arr[index] << endl;
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter 5 elements: 5

8
8
2
1
Enter index: 4
Value: 1
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-9

main.cpp

```cpp
#include <iostream>
using namespace std;

class Demo {
public:
    void operator()(int a) {
        cout << "One argument: " << a << endl;
    }

    void operator()(int a, int b) {
        cout << "Two arguments: " << a << " " << b << endl;
    }
};

int main() {
    Demo d;
    d(5);
    d(5,10);
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
One argument: 5
Two arguments: 5 10
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-10

main.cpp

```cpp
#include <iostream>
using namespace std;

class Test {
    int x;

public:
    friend istream& operator>>(istream &in, Test &t);
    friend ostream& operator<<(ostream &out, Test &t);
};

istream& operator>>(istream &in, Test &t) {
    cout << "Enter value: ";
    in >> t.x;
    return in;
}

ostream& operator<<(ostream &out, Test &t) {
    out << "Value = " << t.x;
    return out;
}

int main() {
    Test t;
    cin >> t;
    cout << t;
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter value: 5
Value = 5done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-11

main.cpp

```cpp
#include <iostream>
using namespace std;

class Test {
    float x;

public:
    Test(float f) {
        x = f;
    }

    void show() {
        cout << x << endl;
    }
};

int main() {
    float f;
    cout << "Enter float: ";
    cin >> f;

    Test t = f;
    t.show();
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter float: 5.5
5.5
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-12

main.cpp

```cpp
#include <iostream>
using namespace std;

class Test {
    float x;

public:
    Test(float f=0) {
        x = f;
    }

    operator float() {
        return x;
    }
};

int main() {
    Test t(5.5);
    float f;
    f = t;
    cout << "Converted: " << f << endl;
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Converted: 5.5
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-13

main.cpp

```cpp
#include <iostream>
#include <cmath>
using namespace std;

class Polar {
public:
    float r, angle;

    Polar(float r1, float a1) {
        r = r1;
        angle = a1;
    }
};

class Cartesian {
    float x, y;

public:
    Cartesian(Polar p) {
        x = p.r * cos(p.angle);
        y = p.r * sin(p.angle);
    }

    void show() {
        cout << "x = " << x << " y = " << y << endl;
    }
};

int main() {
    Polar p(10,5);
    Cartesian c = p;
    c.show();
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
x = 2.83662 y = -9.58924
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

OOPS ASSIGNMENT-8

SOHAM SACHDEVA

1024150035

2O11

FUNCTION TEMPLATE

QUES-1

main.cpp

```cpp
#include <iostream>
using namespace std;
template <typename T>
void swapValues(T &a, T &b) {
    T temp;
    temp = a;
    a = b;
    b = temp;
}
int main() {
    int x, y;
    cout << "Enter two integers: ";
    cin >> x >> y;

    cout << "Before swapping: " << x << " " << y << endl;
    swapValues(x, y);
    cout << "After swapping: " << x << " " << y << endl;

    float a, b;
    cout << "Enter two float numbers: ";
    cin >> a >> b;

    cout << "Before swapping: " << a << " " << b << endl;
    swapValues(a, b);
    cout << "After swapping: " << a << " " << b << endl;
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter two integers: 4
10
Before swapping: 4 10
After swapping: 10 4
Enter two float numbers: 3.5
8.5
Before swapping: 3.5 8.5
After swapping: 8.5 3.5
Done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-2

main.cpp

```cpp
#include <iostream>
using namespace std;
template <typename T>
T findMin(T arr[], int size) {
    T min = arr[0];   // assume first element is minimum

    for(int i = 1; i < size; i++) {
        if(arr[i] < min) {
            min = arr[i];
        }
    }
    return min;
}
int main() {
    int n;
    cout << "Enter number of elements: ";
    cin >> n;
    int arr[100];
    cout << "Enter elements: ";
    for(int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    cout << "Minimum element is: " << findMin(arr, n) << endl;
    float farr[3] = {2.5, 1.2, 3.8};
    cout << "Minimum in float array: " << findMin(farr, 3) << endl;
    cout<<"Done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter number of elements: 4
Enter elements: 4
2
1
3
Minimum element is: 1
Minimum in float array: 1.2
Done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-3

main.cpp

```cpp
#include <iostream>
using namespace std;

template <typename T>
void bubbleSort(T arr[], int size) {
    for(int i = 0; i < size - 1; i++) {
        for(int j = 0; j < size - i - 1; j++) {
            if(arr[j] > arr[j + 1]) {
                T temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}

int main(){
    int n;
    cout<<"Enter number of elements: "<<endl;
    cin >> n;
    int arr[100];
    cout<<"Enter elements: "<<endl;
    for(int i=0;i<n;i++){
        cin >> arr[i];
    }
    bubbleSort(arr,n);
    cout<<"Sorted array: "<<endl;
    for(int i=0;i<n;i++){
        cout<<arr[i]<<" ";
    }
    cout<<endl;
    float farr[5] = {3.2, 1.5, 4.8, 2.1, 0.9};
    bubbleSort(farr, 5);
    cout << "Sorted float array: ";
    for(int i = 0; i < 5; i++) {
        cout << farr[i] << " ";
    }
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter number of elements:
4
Enter elements:
4
43
1
2
Sorted array:
1 2 4 43
Sorted float array: 0.9 1.5 2.1 3.2 4.8 done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-4

main.cpp

```cpp
#include <iostream>
using namespace std;
template <typename T>
int linearSearch(T arr[], int size, T key) {
    for(int i = 0; i < size; i++) {
        if(arr[i] == key) {
            return i;   // return index if found
        }
    }
    return -1;  // not found
}
int main() {
    int n;
    cout << "Enter number of elements: ";
    cin >> n;

    int arr[100];
    cout << "Enter elements: ";
    for(int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    int key;
    cout << "Enter element to search: ";
    cin >> key;
    int result = linearSearch(arr, n, key);
    if(result != -1)
        cout << "Element found at index: " << result << endl;
    else
        cout << "Element not found" << endl;
    float farr[5] = {1.1, 2.2, 3.3, 4.4, 5.5};
    float fkey = 3.3;
    int res2 = linearSearch(farr, 5, fkey);
    if(res2 != -1)
        cout << "Float element found at index: " << res2 << endl;
    else
        cout << "Float element not found" << endl;
cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter number of elements: 4
Enter elements: 4
44
4
2
Enter element to search: 4
Element found at index: 0
Float element found at index: 2
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-5

main.cpp

```cpp
#include <iostream>
using namespace std;
template <typename T>
void process(T a) {
    cout << "Single parameter: " << a << endl;
}
template <typename T>
void process(T a, T b) {
    cout << "Two parameters (same type): " << a << " and " << b << endl;
}
template <typename T1, typename T2>
void process(T1 a, T2 b) {
    cout << "Two parameters (different types): " << a << " and " << b << endl;
}
int main() {
    process(10);
    process(5, 15);
    process(3.5, 2.1);
    process(10, 3.14);
    process('A', 100);
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Single parameter: 10
Two parameters (same type): 5 and 15
Two parameters (same type): 3.5 and 2.1
Two parameters (different types): 10 and 3.14
Two parameters (different types): A and 100
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

CLASS TEMPLATE

QUES-1

main.cpp

```cpp
#include <iostream>
using namespace std;
template <class T>
class Stack {
    T arr[100];
    int top;
public:
    Stack() {
        top = -1;
    }
    void push(T value) {
        if(top == 99) {
            cout << "Stack Overflow" << endl;
            return;
        }
        top++;
        arr[top] = value;
        cout << "Pushed: " << value << endl;
    }
    void pop() {
        if(top == -1) {
            cout << "Stack Underflow" << endl;
            return;
        }
        cout << "Popped: " << arr[top] << endl;
        top--;
    }
    void display() {
        if(top == -1) {
            cout << "Stack is empty" << endl;
            return;
        }
        cout << "Stack elements: ";
        for(int i = top; i >= 0; i--) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
};
```

Output

```
Pushed: 10
Pushed: 20
Pushed: 30
Stack elements: 30 20 10
Popped: 30
Stack elements: 20 10
Pushed: 1.1
Pushed: 2.2
Stack elements: 2.2 1.1
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-2

main.cpp

```cpp
#include <iostream>
using namespace std;
template <class T>
class Queue {
    T arr[100];
    int front, rear;
public:
    Queue() {
        front = -1;
        rear = -1;
    }
    void enqueue(T value) {
        if (rear == 99) {
            cout << "Queue Overflow\n";
            return;
        }
        if (front == -1) front = 0;  // first element
        arr[++rear] = value;
    }
    T dequeue() {
        if (front == -1 || front > rear) {
            cout << "Queue Underflow\n";
            return T();
        }
        return arr[front++];
    }
    void display() {
        if (front == -1 || front > rear) {
            cout << "Queue is empty\n";
            return;
        }
        for (int i = front; i <= rear; i++) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
};
```

Output

```
Queue elements: 10 20 30
Dequeued element: 10
After dequeue: 20 30
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-3

main.cpp

```cpp
#include <iostream>
using namespace std;

template <class T1, class T2>
class Pair {
    T1 first;
    T2 second;

public:
    void setValues(T1 a, T2 b) {
        first = a;
        second = b;
    }

    void display() {
        cout << "First value: " << first << endl;
        cout << "Second value: " << second << endl;
    }
};

int main() {
    Pair<int, float> p1;
    p1.setValues(10, 3.14);
    p1.display();

    cout << endl;

    Pair<char, string> p2;
    p2.setValues('A', "Hello");
    p2.display();
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
First value: 10
Second value: 3.14

First value: A
Second value: Hello
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-4

main.cpp

```cpp
#include <iostream>
using namespace std;
template <class T>
class Arithmetic {
    T a, b;
public:
    void setValues(T x, T y) {
        a = x;
        b = y;
    }
    void add() {
        cout << "Addition: " << a + b << endl;
    }
    void subtract() {
        cout << "Subtraction: " << a - b << endl;
    }
    void multiply() {
        cout << "Multiplication: " << a * b << endl;
    }
    void divide() {
        if (b != 0)
            cout << "Division: " << a / b << endl;
        else
            cout << "Division by zero not allowed" << endl;
    }
};
```

Output

```
Addition: 15
Subtraction: 5
Multiplication: 50
Division: 2

Addition: 7.7
Subtraction: 3.3
Multiplication: 12.1
Division: 2.5
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

QUES-5

main.cpp

```cpp
#include <iostream>
using namespace std;
template <class T>
class Array {
    T arr[100];
    int size;
public:
    void input() {
        cout << "Enter number of elements: ";
        cin >> size;

        cout << "Enter elements: ";
        for(int i = 0; i < size; i++) {
            cin >> arr[i];
        }
    }
    void display() {
        cout << "Array elements: ";
        for(int i = 0; i < size; i++) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
};
int main() {
    Array<int> a1;
    a1.input();
    a1.display();
    cout << endl;
    Array<float> a2;
    a2.input();
    a2.display();
    cout<<"done by Soham Sachdeva(1024150035)"<<endl;
    return 0;
}
```

Output

```
Enter number of elements: 4
Enter elements: 1
2
3

4
Array elements: 1 2 3 4

Enter number of elements: 3
Enter elements: 1
6
7
Array elements: 1 6 7
done by Soham Sachdeva(1024150035)


=== Code Execution Successful ===
```

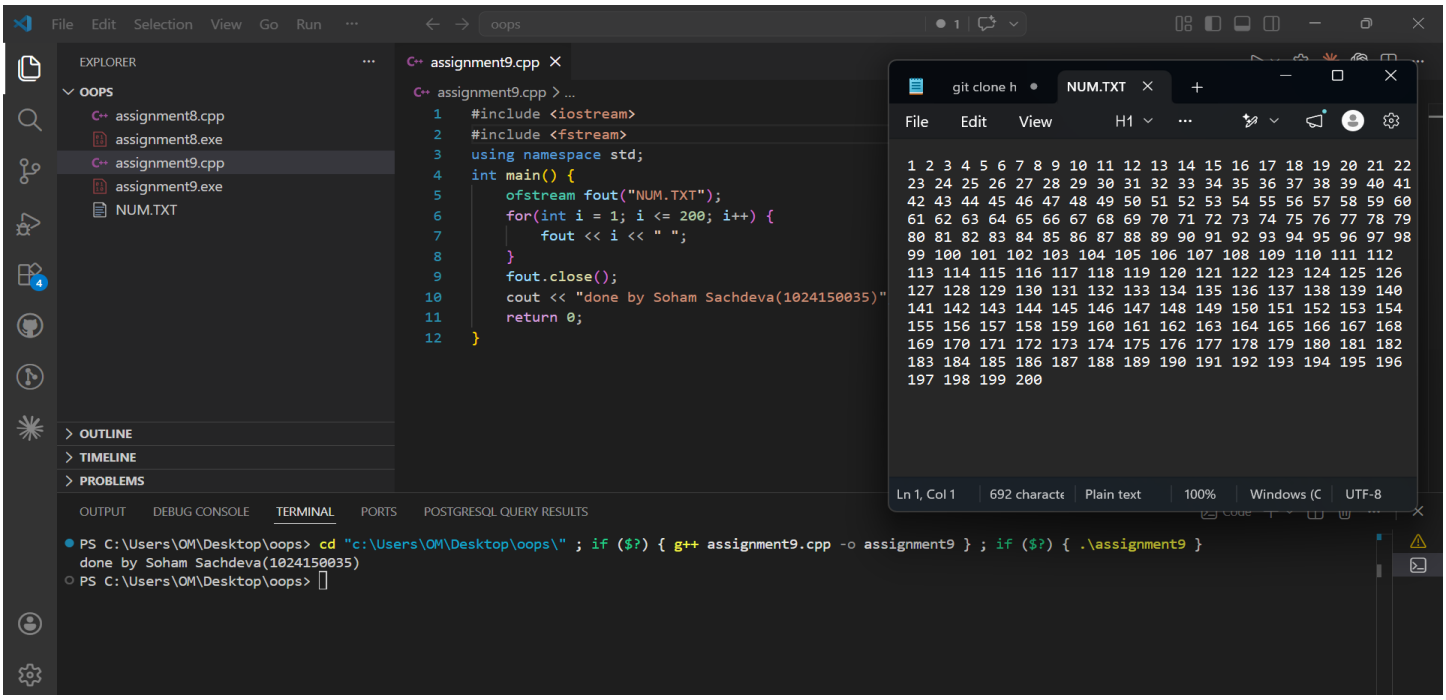
OOPS ASSIGNMENT-9

SOHAM SACHDEVA

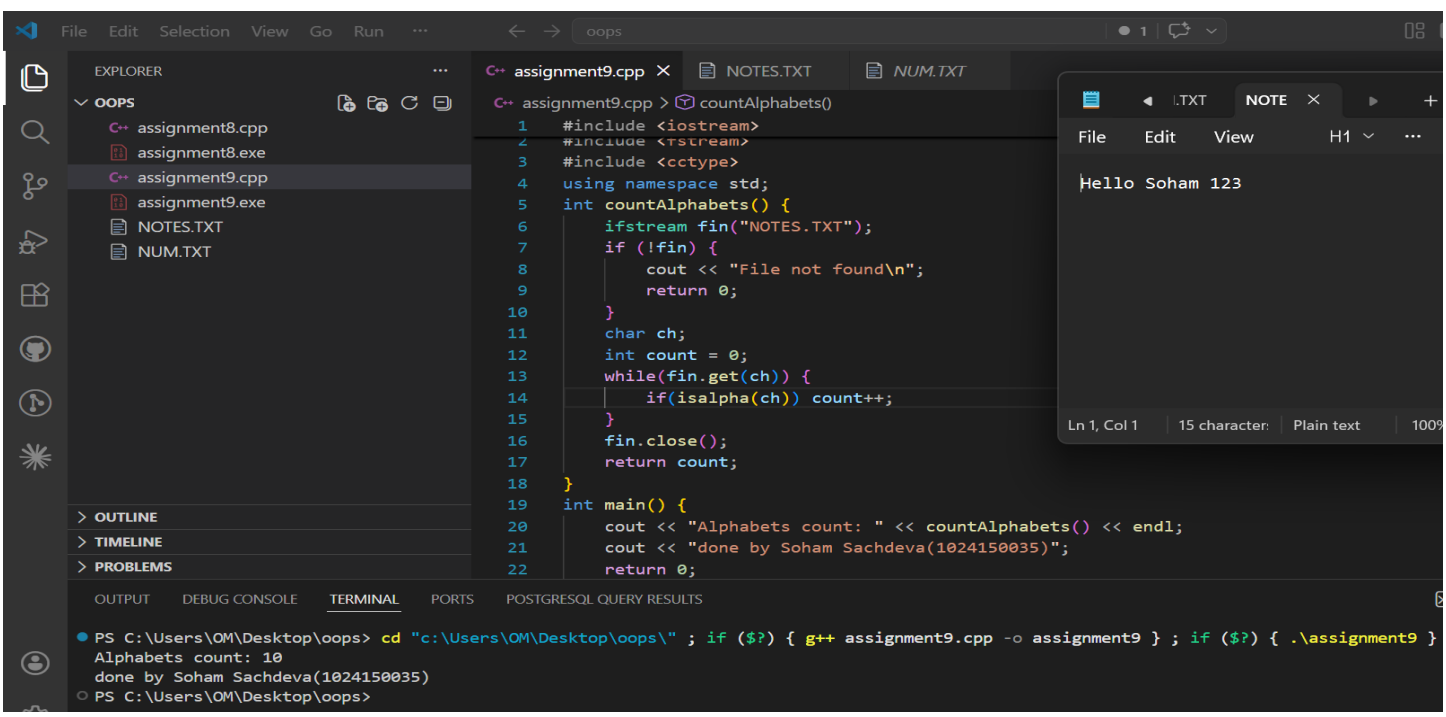
1024150035

2O11

QUES-1



QUES-2



QUES-3

```cpp
#include <iostream>
#include <fstream>
using namespace std;
int main() {
    ifstream fin("source.txt");
    ofstream fout("target.txt");
    char ch;
    while(fin.get(ch)) {
        fout.put(ch);
    }
    fin.close();
    fout.close();
    cout << "File copied successfully\n";
    cout << "done by Soham Sachdeva(1024150035)";
    return 0;
}
```

```
PS C:\Users\OM\Desktop\oops> cd "c:\Users\OM\Desktop\oops\" ; if ($?) { g++ assignment9.cpp -o assignment9 } ; if ($?) { .\assignment9 }
File copied successfully
done by Soham Sachdeva(1024150035)
PS C:\Users\OM\Desktop\oops>
```

QUES-4

```cpp
#include <iostream>
#include <fstream>
using namespace std;
int main() {
    string str;
    cout << "Enter a string: ";
    getline(cin, str);
    cout << "Length: " << str.length() << endl;
    ofstream fout("string.txt");
    fout << str;
    fout.close();
    ifstream fin("string.txt");
    string readStr;
    getline(fin, readStr);
    cout << "Read from file: " << readStr << endl;
    fin.close();
    cout << "done by Soham Sachdeva(1024150035)";
    return 0;
}
```

```
PS C:\Users\OM\Desktop\oops> cd "c:\Users\OM\Desktop\oops\" ; if ($?) { g++ assignment9.cpp -o assignment9 } ; if ($?) { .\assignment9 }
Enter a string: soham
Length: 5
Read from file: soham
done by Soham Sachdeva(1024150035)
```

QUES-5

assignment9.cpp > main()

```cpp
#include <iostream>
#include <fstream>
using namespace std;
int main() {
    // (a) Create file A-Z
    ofstream fout("file.txt");
    for(char ch = 'A'; ch <= 'Z'; ch++) {
        fout << ch;
    }
    fout.close();
    ifstream fin("file.txt");
    // (a) Read 10th character
    fin.seekg(9);
    char ch;
    fin.get(ch);
    cout << "10th character: " << ch << endl;
    fin.close();
    // (b) Overwrite 5th character
    fstream file("file.txt", ios::in | ios::out);
    file.seekp(4);
    file.put('X');
    file.close();
    // (c) File size
    ifstream fin2("file.txt");
    fin2.seekg(0, ios::end);
    cout << "File size: " << fin2.tellg() << endl;
    fin2.close();
    // (d) Last character
    ifstream fin3("file.txt");
    fin3.seekg(-1, ios::end);
    fin3.get(ch);
    cout << "Last character: " << ch << endl;
    fin3.close();
    // (e) data.txt operations
    ofstream f("data.txt");
    f << "Hello\nThis is a test file\nFor file handling";
    f.close();
    ifstream f2("data.txt");
    f2.seekg(10);
    cout << "Position: " << f2.tellg() << endl;
    string line;
    while(getline(f2, line)) {
        cout << line << endl;
    }
    f2.close();
    cout << "done by Soham Sachdeva(1024150035)";
    return 0;
}
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTGRESQL QUERY RESULTS

```
PS C:\Users\OM\Desktop\oops> cd "c:\Users\OM\Desktop\oops\" ; if ($?) { g++ assignment9.cpp -o assignment9 } ; if ($?) { .\assignment9 }
10th character: J
File size: 26
Last character: Z
Position: 10
s is a test file
For file handling
done by Soham Sachdeva(1024150035)
PS C:\Users\OM\Desktop\oops>
```

QUES-6

assignment9.cpp × target.txt NOTES.TXT

assignment9.cpp > main()

```cpp
#include <iostream>
#include <fstream>
using namespace std;
int main() {
    fstream file("hello.txt", ios::out | ios::in);
    string str = "Hello World";
    for(int i = 0; i < str.length(); i++) {
        file.put(str[i]);
        cout << "Position: " << file.tellp() << endl;
    }
    // Replace "World" with "C++"
    file.seekp(6);
    file << "C++";
    file.close();
    cout << "done by Soham Sachdeva(1024150035)";
    return 0;
}
```

OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTGRESQL QUERY RESULTS

```
PS C:\Users\OM\Desktop\oops> cd "c:\Users\OM\Desktop\oops\" ; if ($?) { g++ assignment9.cpp -o assignment9 } ; if ($?) { .\assignment9 }
Position: -1
Position: -1
Position: -1
Position: -1
Position: -1
Position: -1
Position: -1
Position: -1
Position: -1
Position: -1
Position: -1
done by Soham Sachdeva(1024150035)
PS C:\Users\OM\Desktop\oops>
```